

A Smart Twist to the Power Grid

Whether the coming generations will have enough energy to power lights, or have to resort to oil lamps and wax candles depends on how smart our power grids become—and fortunately they are gearing up quite fast, in cohort with the Internet of Things, to manage energy requirements smartly

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Despite the fact that the current generation of devices and appliances, including lights and fans, are smart and capable of conserving power to a large extent, there is still a clear shortage of power. The issue is discussed widely in newspapers and manifested inconveniently as long power outages. What is the way out?

"The smart grid is a concept that will keep all the lights on not only for this generation but the next generation as well. It would not only give us independence in our energy requirements but will also enable a reliable source to keep our ever-increasing energy demands in control. Technically, a smart grid is a bi-directional grid which facilitates an integration of all players in the energy ecosystem in a reliable, economical and sustainable model," says Reddy Karri, executive vice president, Symphony Teleca Corp.

Satish Mohanram, technical marketing manager, National Instruments, India, explains, "A smart grid is termed so because it can understand the load and supply behaviour and manage the same in the most optimised fashion with very less or even no manual intervention. It delivers higher performance, lower cost and essential new features such as seamless integration of loads, grid-level energy storage and distributed renewable energy generation."

The smart grid, networked grid or digital grid primarily aims at balancing limited power-production capacity and increasing demand for energy, by re-



Basic view of a smart grid

ducing losses, increasing efficiency, optimising the demand distribution and integrating small and large renewable energy sources like solar and wind. "Structurally speaking, the smart grid is a network between the generation plant, transmission and distribution stations, which in turn extends to smart meters in homes, businesses or factories," says Shailesh Thakurdesai, business development manager, Micro-controllers, Texas Instruments (India).

Making a grid smart

"In my opinion, the scope of smart grid is very vast, as there are many details that impact design from grid infrastructure to end nodes. To deploy a smart grid, it would be very important to first understand and establish the scope of the project. Predominantly, the scope of a smart grid project will aim to achieve some of the following: reduction of distribution losses, achieving bi-directional response for

real-time control, increasing overall capacity, managing and controlling for peak demand and multi-tariff implementation. Based on this objective, one can proceed to evaluate the current technology and topology to be used. On assessing this, a pilot project can be conducted to see the actual results of the smart grid," says Thakurdesai.

Typically, the evolution of a smart grid happens organically across four areas: generation, transmission, distribution and retail.

At the generation level, one of the main benefits of the smart grid is that it enables distributed energy production. Theoretically, it enables even individuals to be contributors to the power grid. If you have a solar panel on your terrace and are producing more power than you consume, you can contribute that to the grid and get a discount on your next bill. Although that is still not very common, the smart grid has definitely made it possible to